

GraceEMO Control Stack

Body analogy mapped onto this repository. The brain decides what to do. The kernel routes events. Motor control already exists as differential drive on /cmd_vel. Do not train walking.

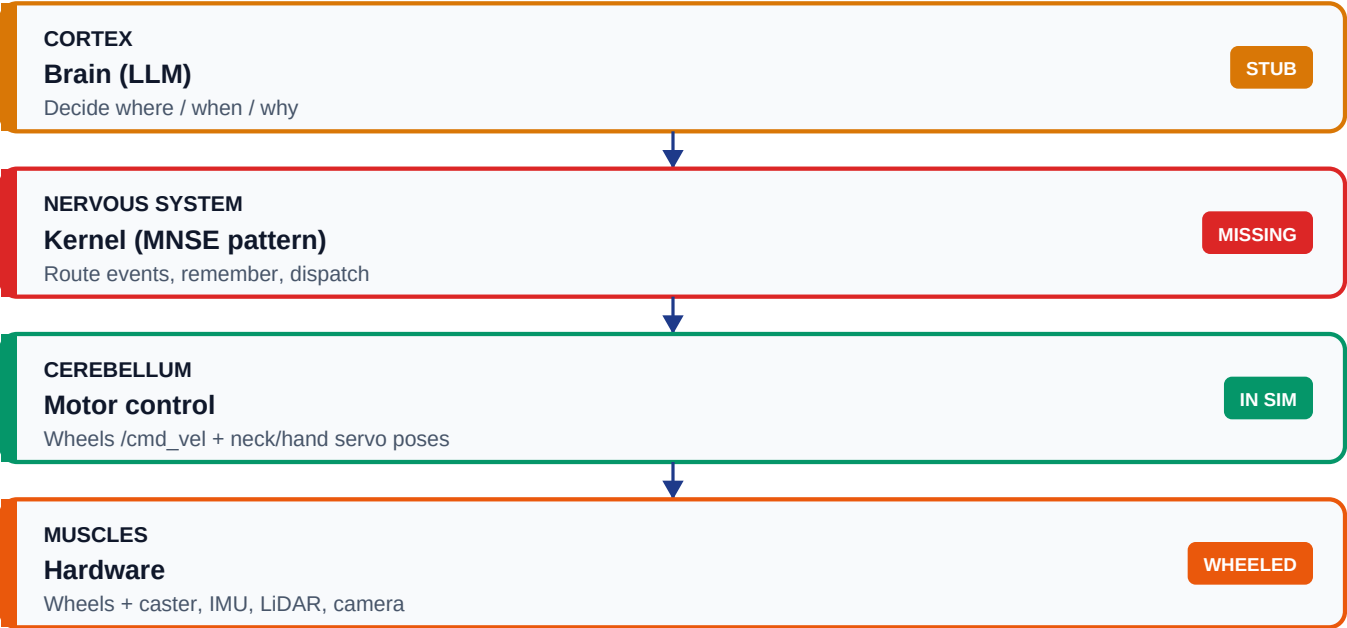
Stub Cortex — llm_node	Missing Nervous system — kernel	Done Cerebellum — /cmd_vel	Wheels Hardware in this sim
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Bottom line. You do not train walking. This sim already moves. The job is a brain that knows **where, when, and why** — then a one-line command to the existing motor layer. A future LPU humanoid swaps only that last hop.

Phase 1 lock — wheeled base + neck + hands. Drive with wheels on /cmd_vel. Expression is neck look-at and simple hand poses ({hand-hi}, {hand-up}, {hand-down}) — not biped gait or full-arm inverse kinematics.

1. Four layers, one command path

Cortex and kernel are the team's work. Cerebellum and hardware are already provided in simulation. Decision flows downward; sensors and completions flow up.



What each layer handles

Layer	Handles	Does not handle	In this repo
Brain (LLM)	Intent, dialogue, task choice	Joint angles, gait timing	llm_node.py keyword stubs; Gemini in yaml, not called
Kernel	Events, memory, dispatch	IMU balance at 500 Hz	planner_node.py if/else; no event bus or InspectorState
Motor (/cmd_vel + servos)	Drive, look_at, named hand poses	Biped gait, full-arm IK, ZMP balance	Drive exists; neck/hands not in URDF yet
Hardware	Wheels now; neck + 90° hands in Phase 1	Walking legs	gracemo.urdf.xacro is wheeled only; legacy Pi servo in functions.py

Human analog: cortex → nervous system → cerebellum → muscles. Same split as MNSE (nervous system) vs robot firmware (cerebellum).

2. Designed architecture vs running code

brain.txt describes Option B: a Rust kernel plus Python adapters. This workspace shipped Option A: everything in ROS 2 Python. The brain is still a stub.

Designed (brain.txt) EventBus, InspectorState, SQLite ledger, knowledge graph, action dispatcher, MCP, K-language behaviors. Target demo: “Hey GraCemo, go to the door” → memory recall → navigate_to → NavigationComplete → speak.	Running (this repo) ROS 2 nodes: vision, voice, memory package, llm_node, planner_node, Virtual Space sim. Planner greets on person detect, stops on “stop”, otherwise asks the LLM stub. No waypoint dispatch from voice.
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Component checklist

Piece	MNSE note	GraceEMO-Final
Event bus	Reuse / build	Not present
Ledger + graph	SQLite in kernel	gracemo_memory exists, not the dispatcher
InspectorState	Live pose, battery, person	Scattered on topics
Action dispatcher	navigate_to, speak, stop	Partial stop + greet
LLM	Gemini for planning	If/else strings in llm_node
ROS 2 bridge	New Python adapter	Nodes are already ROS 2
Motor / walking	Manufacturer firmware	Wheeled /cmd_vel in sim
Gazebo / studio	Needed for demo	Virtual Space on port 8888

Do not import MNSE into this tree by default. B3-lite fits: keep ROS 2 as the bus, add a thin kernel in Python (or Rust later) in front of /cmd_vel. Forking MNSE (B1) or robot-mode MNSE (B2) only makes sense if MNSE is an official dependency and the mentor accepts a Rust desktop kernel as the robot brain.

3. Project scope — build vs skip

Reinforcement learning for biped walking is a multi-month research track. This campus demo does not need it. The sim already accepts high-level motion the same way a prebuilt humanoid would.

Effort that belongs in this project (walking RL excluded)



- Done 35% — sim, URDF, topics
- Build 55% — kernel, LLM, memory actions
- Later 10% — swap wheels for LPU humanoid API

Build	Why	Skip	Why skip
InspectorState	One live picture of the robot	Train walking (PPO / SAC)	Phase 1 is wheeled; walking is later
Event bus	Vision, voice, odom in one place	Balance / ZMP at 500 Hz	Not needed on a wheeled base
Memory recall	Door is at a known pose	Full-arm inverse kinematics	Hands are 90° named poses, not pick-and-place
navigate_to / stop / speak	Wheels: kernel says what; /cmd_vel does how	Fall recovery motions	Out of campus-demo scope
look_at + hand poses	Neck pan/tilt; {hand-hi} {hand-up} {hand-down}	Isaac Sim walking policy	GPU + months; sim-only RL can be paper future work
Wire Gemini	Cortex is still hardcoded replies	Google Maps as occupancy world	Terms of service + wrong data type for Nav2

If LPU later provides a walking humanoid

Keep cortex and kernel. Phase 1 locomotion stays wheeled. Replace only the last hop later: `navigate_to` still fires, but the base becomes `robot.walk(...)` instead of `/cmd_vel`. Neck and hand actions stay the same.

Phase 1 motor surface (what the kernel is allowed to send)

Action	Hardware	Notes
navigate_to / stop	Wheels	Already /cmd_vel in Virtual Space
look_at / look_home	Neck pan + tilt	Not in URDF yet; 2 servos is enough
hand_hi / hand_up / hand_down	Hands, 90° only	Already specified in data/8-end-functions.txt
speak	TTS / speaker	Existing /gracemo/say

Ask LPU before writing a hardware bridge

1. Does the robot already walk or drive on its own?
2. Commands: walk-forward API, joint angles, or ROS 2 `/cmd_vel`?
3. Published sensors: IMU, camera, LiDAR, battery, joint states?

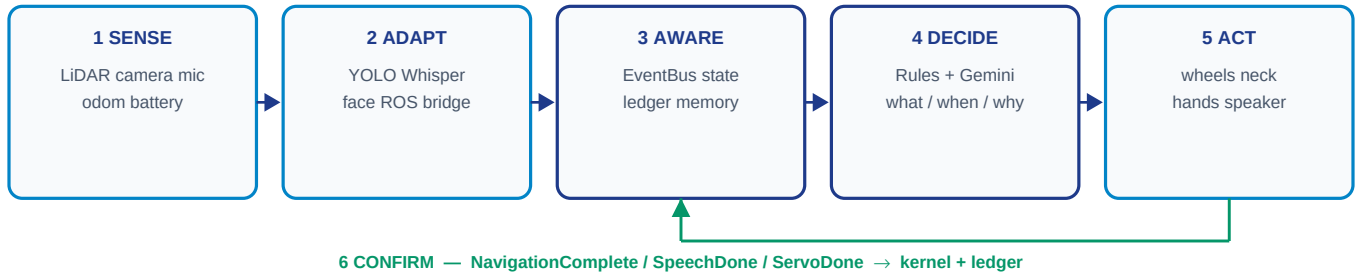
Until then, Virtual Space is the manufacturer controller: Twist in, odom / scan / camera out.

Next implementation slice

Status	Work
Pending	InspectorState: pose, battery, obstacle, person, last voice command
Pending	Dispatcher: navigate_to, stop, speak, look_at, hand_hi / hand_up / hand_down
Pending	Phase 1 hardware: add neck (pan/tilt) and 90° hand joints to URDF; keep wheeled base
Pending	Named places (door, lab) recalled from memory, not hardcoded strings only
Pending	Voice “go to the door” → kernel → patrol / Nav2 waypoint
Pending	Replace llm_node heuristics with Gemini (already in brain.yaml)
Out of scope	Do not start MuJoCo walking, ZMP, or full-arm IK

4. Full Phase 1 flow (closed loop)

Nothing is fire-and-forget. Sensors enter the kernel as events. The kernel updates live state, consults memory, decides, and dispatches named actions. Wheels, neck, hands, and speech execute. Completions come back so the robot can speak, log, and take the next step. Walking is not in this loop.



The three kernel loops (MNSE pattern)

Loop	Budget	Question	Phase 1 example
Perception	< 10 ms	What is happening right now?	InspectorState: pose, battery, obstacle_ahead, person_visible, last_voice
Memory	< 100 ms	What happened before? What is related?	Ledger row + graph: door=(x,y); this person was greeted 20s ago
Reaction	< 500 ms	When X happens, do Y	obstacle < 1 m → stop; "go to the door" → navigate_to; person close → greet

Events in (adapters → kernel)

Event	ROS source	Kernel field	Today
LaserScan	/scan	obstacle_ahead, min_range	Published by Virtual Space
Odometry	/odom	position, yaw, speed	Published by Virtual Space
IMU	/imu/data	orientation (optional)	Published by Virtual Space
RGB image	/camera/image_raw	(via vision adapter)	Synthetic camera in sim
Detection	/gracemo/detections	person_visible, label, distance	YOLO node exists
VoiceCommand	/gracemo/voice_command	last_voice, intent, entities	Voice package exists
RobotState	/gracemo/robot_state	battery, task, status	Partial in virtual_space_node
Remember / recall	/gracemo/remember	named places, people	memory_node service exists

Actions out (kernel → Phase 1 body)

Action	Goes to	ROS / hardware	Today
stop()	Wheels	/cmd_vel linear=0 angular=0	planner_node on “stop”
navigate_to(name x,y)	Wheels + Nav2	NavigateTo action or /goal_pose	patrol_node has waypoints; not voice-wired
look_at(person point)	Neck pan/tilt	joint_trajectory or servo topic	Not in URDF
look_home()	Neck	servos to forward	Not in URDF
hand_hi / hand_up / hand_down	Hands 90°	named pose → 1-DOF joints	Protocol in 8-end-functions.txt only
speak(text)	Speaker	/gracemo/say and SpeakText action	planner publishes /gracemo/say
express(emotion)	Face / TTS tag	<exp-happy> with speech	Legacy protocol; not in ROS nodes

Completions back (body → kernel)

Completion	Meaning	Next kernel step
NavigationComplete / NavigateTo result	Arrived or failed	speak; look_at if person; ledger; next goal or idle
SpeechDone / SpeakText result	Utterance finished	listen again, or continue task
ServoDone (neck / hands)	Pose reached	allow next gesture; do not stack conflicting poses
Collision / obstacle < 1 m	Unsafe to continue	stop(); cancel Nav2; speak warning; wait or replan
Battery low	Cannot finish patrol	behavior tree: ReturnToCharger (already in main_tree.xml)

ROS 2 map for this loop

Name	Type	Role in the flow
/cmd_vel	geometry_msgs/Twist	Wheeled cerebellum — only locomotion command in Phase 1
/odom /scan /imu/data /camera/image_raw	sensor / nav msgs	Raw sense from Virtual Space (or later hardware)
/gracemo/detections	gracemo_interfaces/Detection	Vision adapter output
/gracemo/voice_command	gracemo_interfaces/VoiceComm and	STT + intent into the kernel
/gracemo/say	std_msgs/String	TTS input
/gracemo/ask_question	AskQuestion service	Cortex: question + context → answer, intent, suggested_actions
/gracemo/remember	Remember service	Write episodic / semantic facts
NavigateTo	action	Long-running drive to a named pose
SpeakText	action	Long-running speech with progress

5. Worked scenarios (every hop)

These are the Phase 1 demos. Each row is one hop. Green exists in the repo, amber is partial, red is not wired yet.

Scenario A — Person appears: stop, look, wave, greet

#	Hop	Who	Today
1	Camera frame at 15 Hz	Virtual Space → /camera/image_raw	Works
2	YOLO: person, 2.1 m, confidence 0.8	detector_node → /gracemo/detections	Node exists
3	InspectorState.person_visible = true	Kernel perception loop	Missing — planner has a local if
4	Cooldown: last greet > 15 s	Memory / kernel	Hardcoded 15 s in planner_node only
5	stop() so the base does not roll into them	Dispatcher → /cmd_vel	Not on greet; only on “stop”
6	look_at(person) using detection bearing	Neck pan/tilt	No neck joints
7	hand_hi (90° wave)	Hand servo named pose	Tag exists in 8-end-functions.txt; no ROS
8	speak “Hello! I am GraceEMO...”	/gracemo/say → TTS	planner_node does this
9	SpeechDone; ledger: greeted track_id	Kernel memory loop	Not stored
10	Listen, or resume idle / patrol	Reaction loop	No return-to-task

Scenario B — “Hey GraCemo, go to the door” (complete)

#	Hop	Who	Today
1	Wake word + STT: “go to the door”	voice → /gracemo/voice_command	Package exists; intent unused
2	Kernel sets last_voice; asks cortex	EventBus + AskQuestion	planner calls LLM only as fallback speak
3	Gemini: intent NAVIGATE, entity door	llm_node / Gemini	Keyword stubs; Gemini not called
4	Memory recall: door → pose or waypoint name	/gracemo/remember + graph	Waypoints only inside patrol_node
5	look_home(); hand_down() travel pose	Neck + hands	Not in URDF
6	navigate_to(location_name=door)	NavigateTo action	Interface exists; nobody sends the goal
7	Wheels follow /cmd_vel; LiDAR in the loop	Virtual Space / Nav2	Drive works in studio
8	If /scan < 1 m: stop, cancel goal, speak	Safety reaction	Sim collides and zeros linear_v only
9	NavigationComplete (success, distance, time)	Action result → kernel	Not wired
10	look_at(person) if someone at the door, else look_home	Neck	Not in URDF
11	speak “I’ve arrived at the door”	TTS	Not triggered
12	Ledger: trip door, success, timestamp	Memory	Not stored

Scenario C — Safety always wins

Trigger	Kernel reaction	Body
User says “stop” / “halt”	Cancel NavigateTo; stop(); speak confirm	Wheels zero — this path exists
LiDAR hit < 1 m while moving	emergency_stop; do not wait for LLM	Sim already blocks penetration
Battery low (behavior tree)	ReturnToCharger over greet/patrol	XML exists; nodes not implemented
Teleop /cmd_vel vs kernel	Safety multiplexer: stop always highest priority	Studio and planner both publish /cmd_vel

Behavior tree (already in-repo sketch)

config/behavior_trees/main_tree.xml is the high-level Fallback: **safety first** (battery → charger), then **greet** (human → stop/face → speak), else **patrol** (next waypoint → navigate). Phase 1 adds look_at + hand_hi on greet, and look_home + hand_down on patrol. The tree does not walk.

6. What “done” looks like for Phase 1

Demo	Must happen end-to-end
Greet	See person → stop wheels → look → wave → speak → remember
Navigate	Hear “go to the door” → recall pose → travel pose → drive → arrive → speak
Safety	Say “stop” or see obstacle → motion halts before the LLM replies
Idle	No person, no command → patrol waypoints with hands down, neck forward

Mentor one-liner

ROS 2 is the bus. The kernel is the closed loop. Sense → remember → decide → wheels / neck / hands / speech → confirm. Phase 1 is a wheeled campus greeter, not a walking humanoid. We do not train gait; we finish this flow.

Sources: brain.txt; planner_node.py; llm_node.py; virtual_space_node.py; gracemo.urdf.xacro; gracemo_interfaces (Detection, VoiceCommand, RobotState, NavigateTo, SpeakText, AskQuestion, Remember); main_tree.xml; data/8-end-functions.txt. Status of this workspace, not a live robot.